E-P.I.P.

Environmental and Personal Information Processor

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Table of Contents

I.
INTRODUCTION

Introductory Overview of *E-P.I.P.*

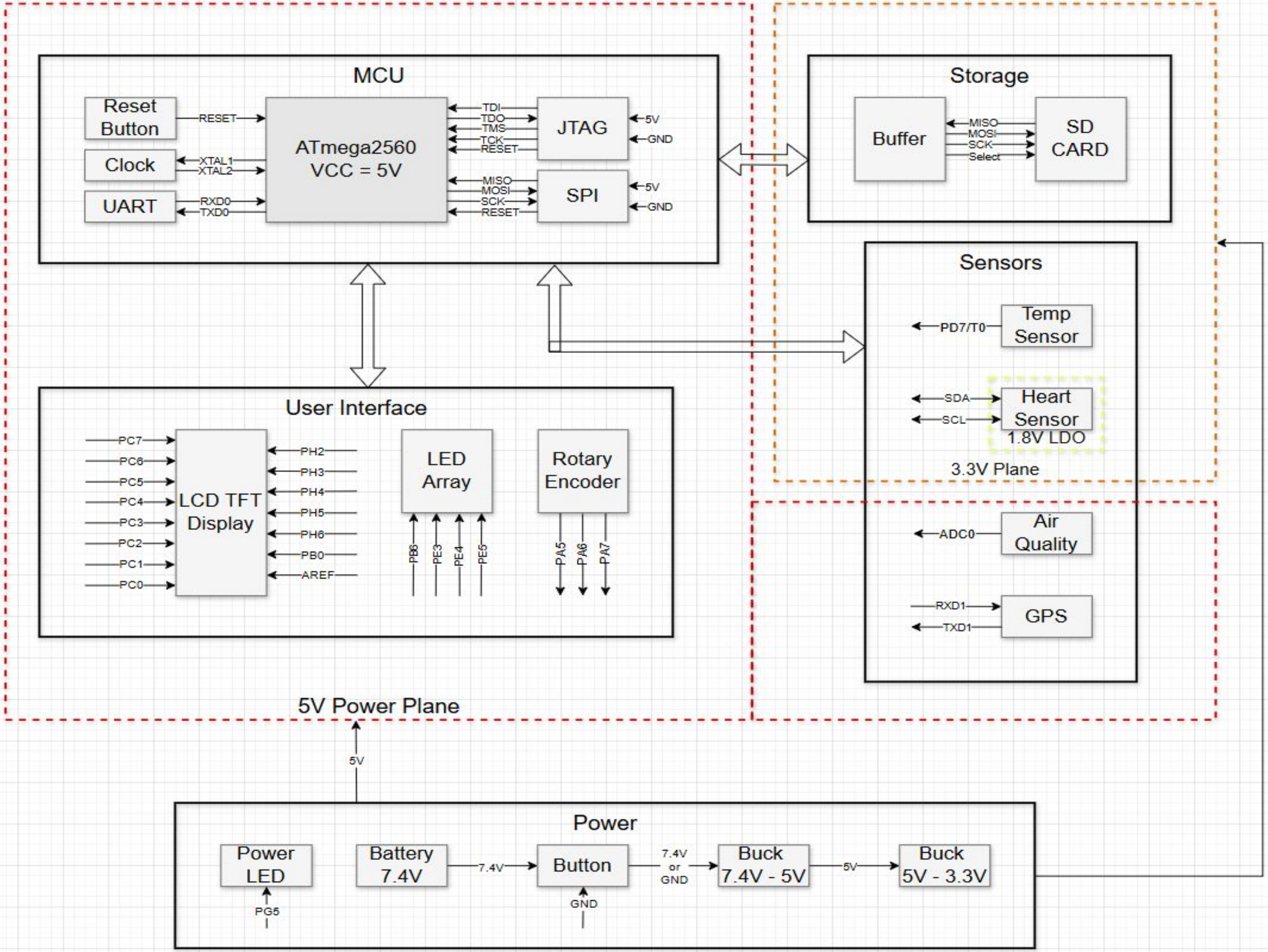
<< Main Purpose >>

Safety device designed for harsh environmental conditions to **keep users informed and safe** by providing crucial data in critical situations.

<< Key Functions >>

1. Temperature & Humidity
2. Air Quality
3. Location
4. Vitals
5. Store Data

E-P.I.P. Block Diagram



II. MCU

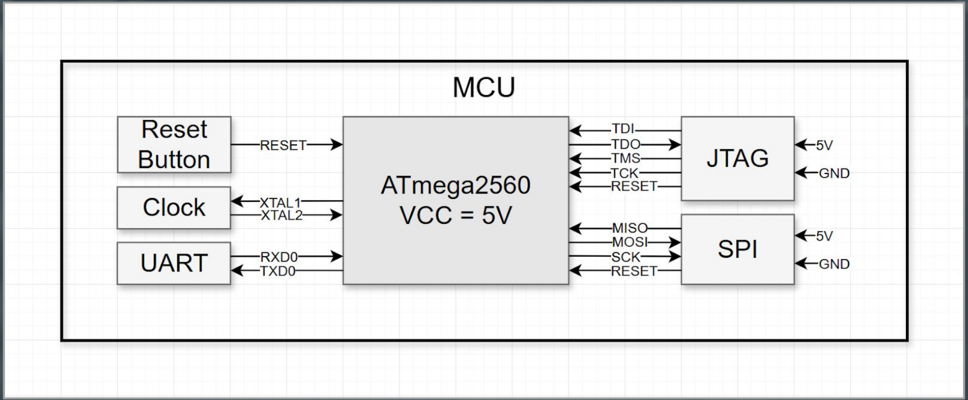
MCU: Overview

Purpose & Requirements:

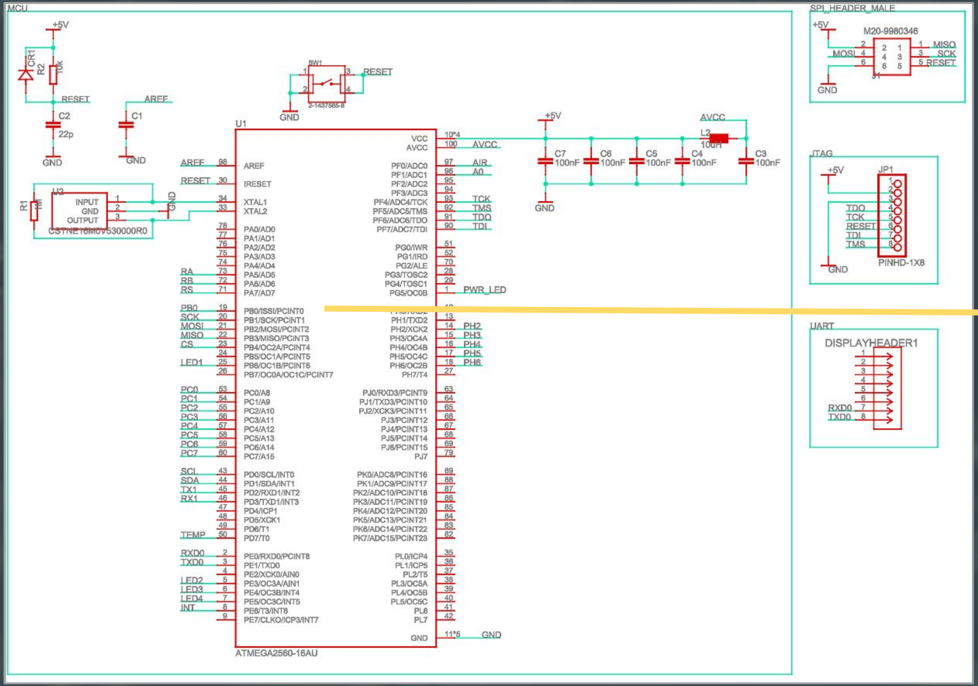
- Sensor data collection
- Input for user interface
- Display sensor readings

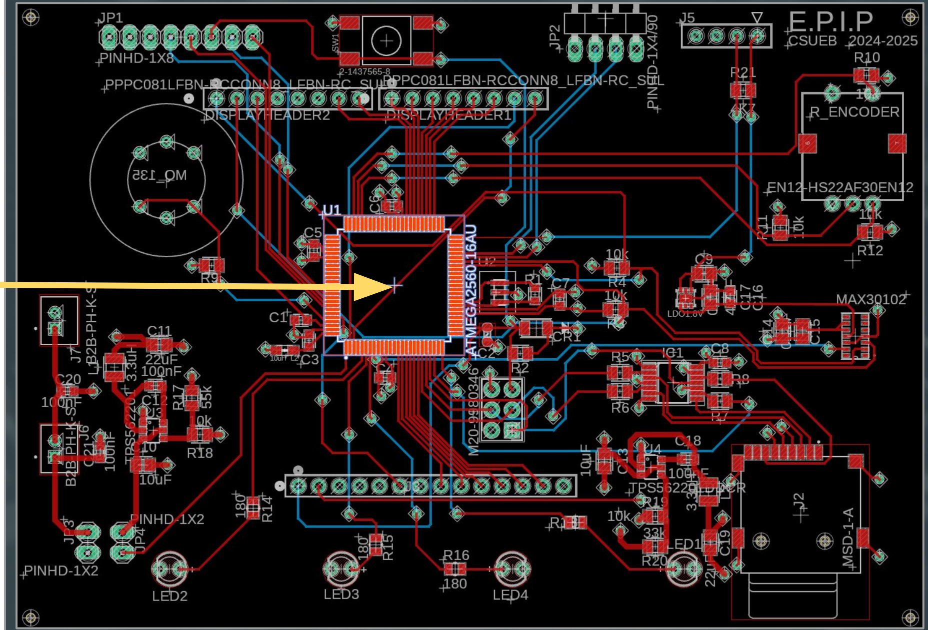
Part(s):

- ATmega2560
- 16MHz clock
- Reset Button
- JTAG, SPI, UART headers



MCU: Schematic & Layout





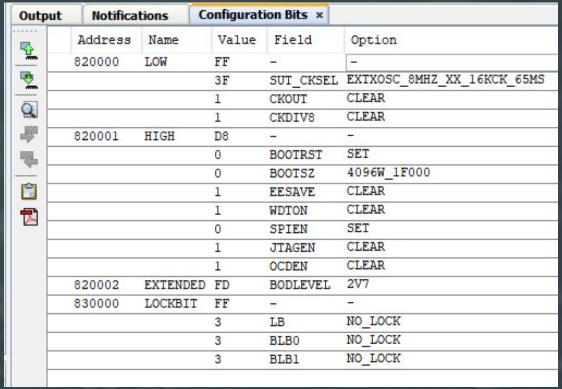
Alberto R.

MCU: Programming

Bootloader compiled and exported from Arduino IDE as .hex file

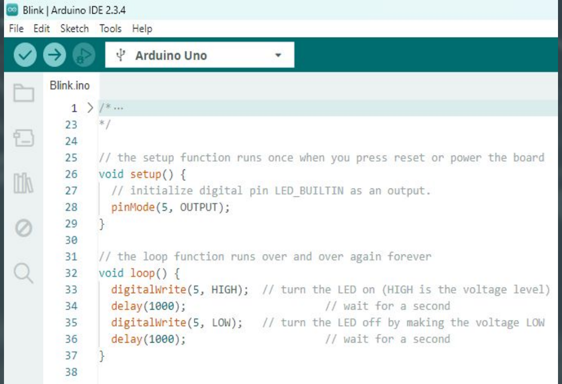
Configuration bits set via MPLAB X IDE:

- LOW: 0xFF → Set external clock and No changes to timing
- HIGH: 0xD8 → SPIEN + BOOTRST, JTAG disabled
- EXTENDED: 0xFD → BODLEVEL = 2.7V
- Lock bits: 0xFF → No memory locking



Output | Notifications | Configuration Bits

Address	Name	Value	Field	Option
820000	LOW	FF	-	-
		3F	SUT_CKSEL	EXTXOSC_8MHZ_XX_16KCK_65MS
		1	CKOUT	CLEAR
		1	CKDIV8	CLEAR
820001	HIGH	D8	-	-
		0	BOOTRST	SET
		0	BOOTSZ	4096W_1F000
		1	EESAVE	CLEAR
		1	WDTON	CLEAR
		0	SPIEN	SET
		1	JTAGEN	CLEAR
		1	OCDEN	CLEAR
820002	EXTENDED	FD	BODLEVEL	2V7
830000	LOCKBIT	FF	-	-
		3	LB	NO_LOCK
		3	BLB0	NO_LOCK
		3	BLB1	NO_LOCK



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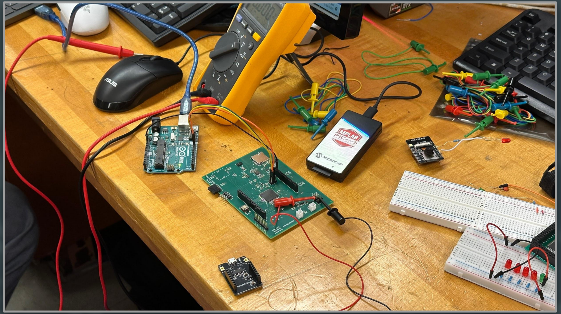
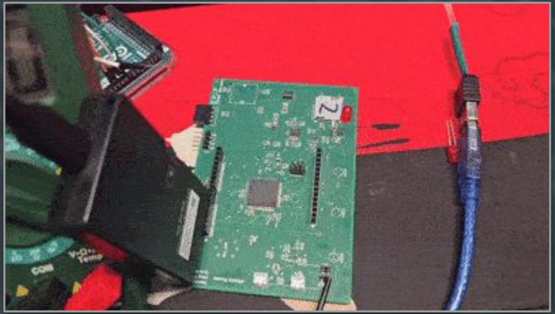
MCU: Milestones

Milestone 1: Verified stable power domains, but crystal oscillator failed { SPI programming unsuccessful }
Milestone 2: Bootloader successfully flashed via JTAG using MPLAB IPE and PICkit 5
Milestone 3: Transitioned to UART uploads using a modified Arduino UNO as USB-to-Serial converter





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III. Temperature & Humidity

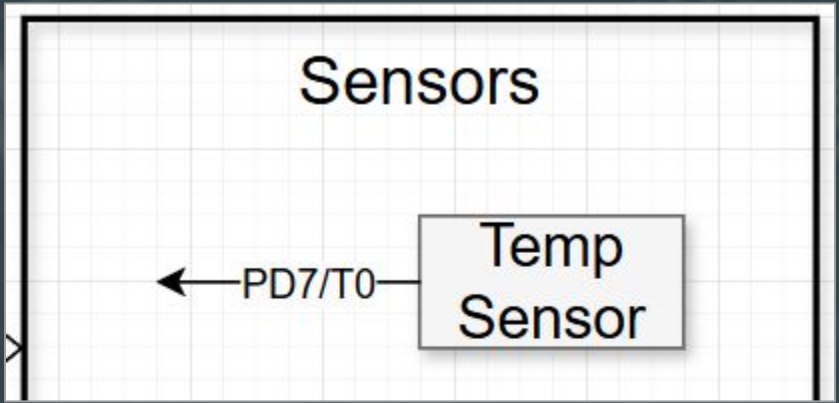
🌡️ Temperature & Humidity: Overview

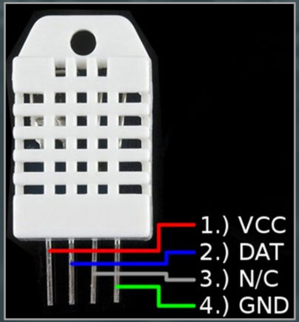
📝Requirements:
- Read temperature
- Read humidity
- Store reading

</> Software:
- Sensor reading
- Data storing

🔗Integration:
- Connect via female header pin
- Top right of the PCB

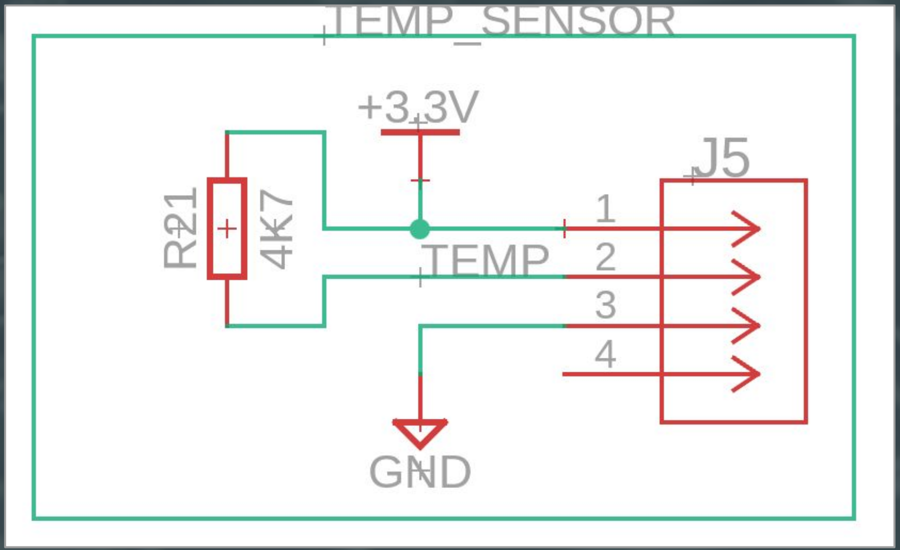
🚩 Reach Goals:
- Read without delay

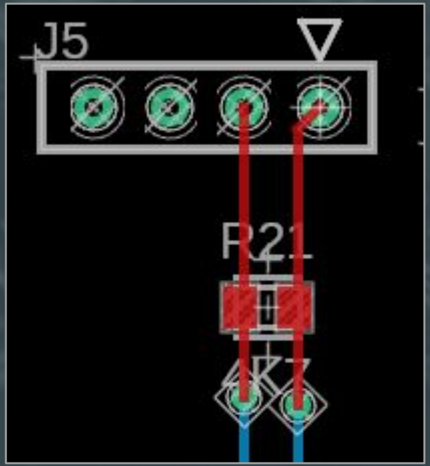




Cody H.

🌡️ Temperature & Humidity: Schematic & Layout





Cody H.

🌡️ Temperature & Humidity: Milestones

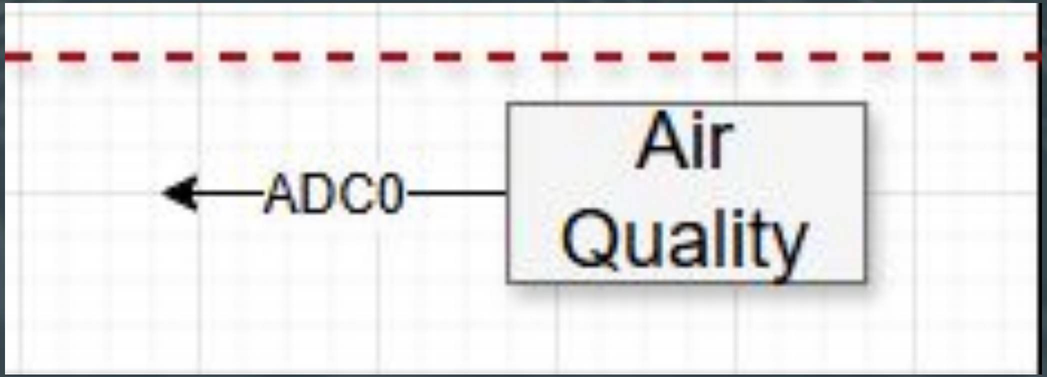
Milestone 1: No Issue; Able to switch on and read temp/humidity	
Milestone 2: Improvement; - Made it read as fast as possible: 2 seconds; likely hardware issue - Met all core functional goals	

Cody H.

☁️ IV. Air Quality

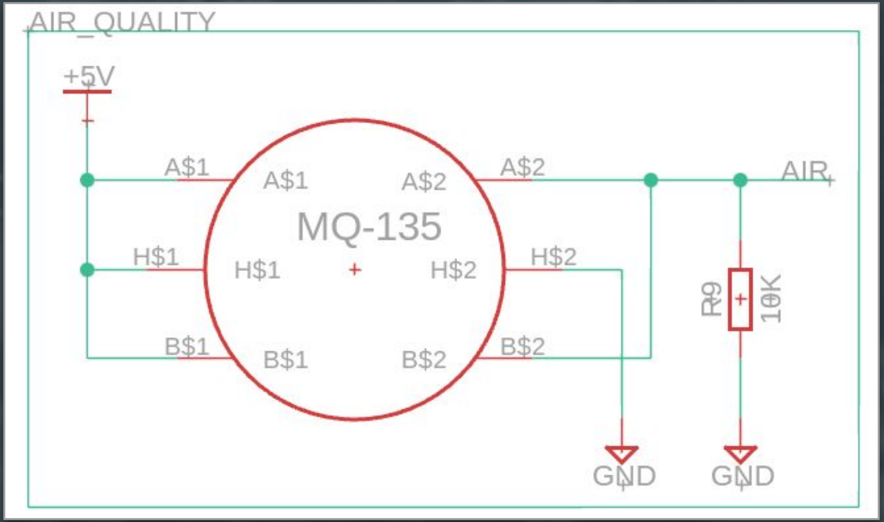
☁️ Air Quality: Overview

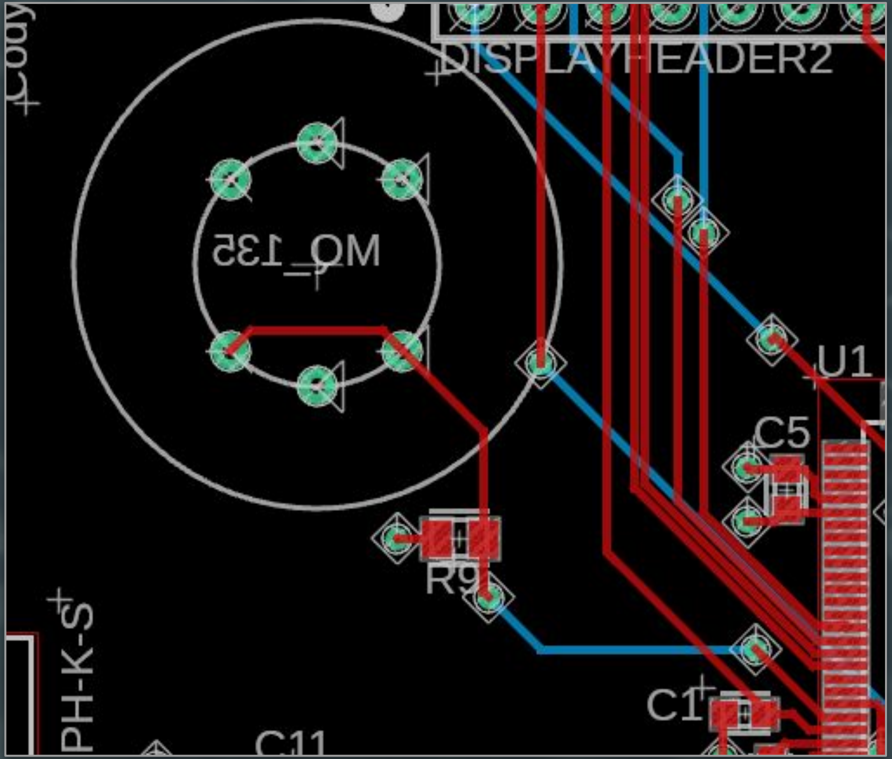
📝Requirements:	</> Software:	🔗Integration:	🚩 Reach Goals:
• Read/detect harmful gases (PPM) • Store readings	• Sensor reading • Data storing	• Connect onto back of PCB • Soldered on	• Change layout footprint



Cody H.

☁️ Air Quality: Schematic & Layout





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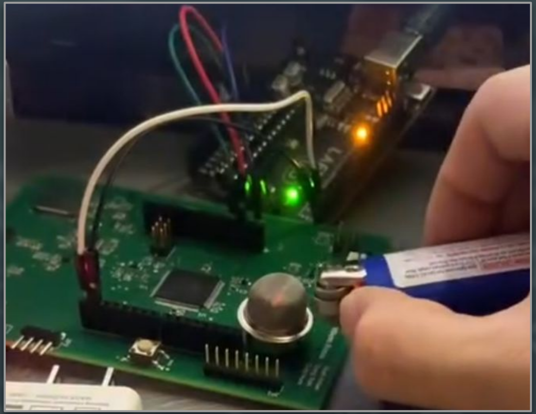
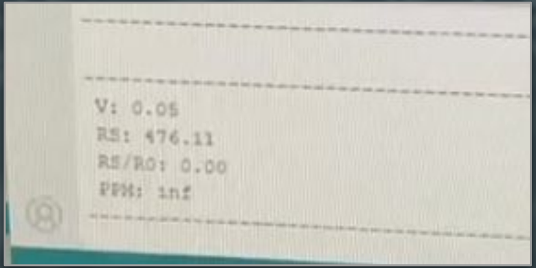
☁️ Air Quality: Milestones

Milestone 1: Had to wire because of sizing; - Read air quality
Milestone 2: Shaved Sensor pins to solder onto PCB - Read and read air quality - Store them to be called on - Fast effective readings





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V. GPS

🛰️ GPS: Overview

📝Requirements:

- NEO-6M GPS module w/ external antenna
- 5V input

</> Software:

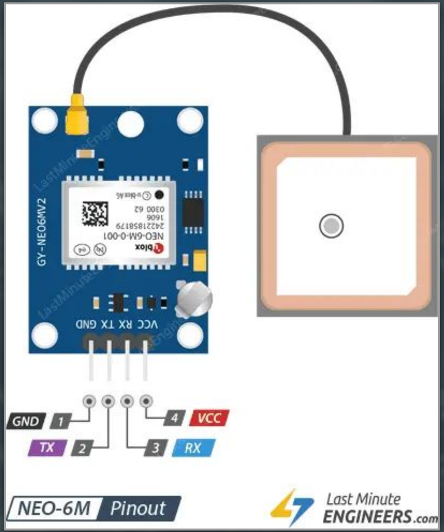
- Communication with MCU via UART
- TinyGPSPlus library
    - Locate real-time position
    - Find #'s of satellites
    - Get time

🔗Integration:

- Connection to MCU via RXD1 and TXD1

🚩 Reach Goals:

- Time sync without satellite coverage



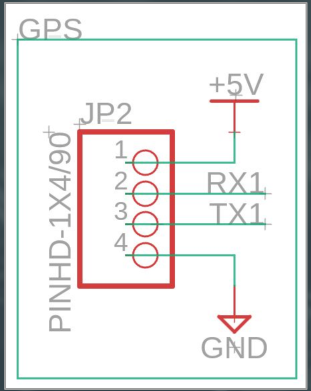
🛰️ GPS: Schematic & Layout

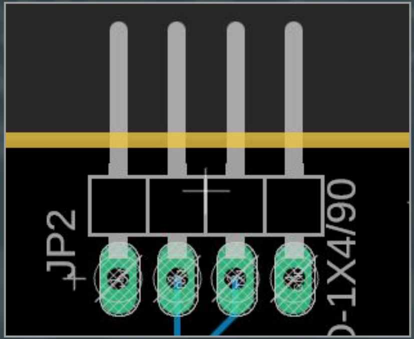
Schematic:

- Pin 1 (5V) connects to VDD
- Pin 2 (RX1) connects to RX (receiver) pin
- Pin 3 (TX1) connects to TX (transmitter) pin
- Pin 4 connects to GND

Layout:

- Right angle 1x4 male pin header connects to the NEO-6M GPS module via female-to-female pin connection



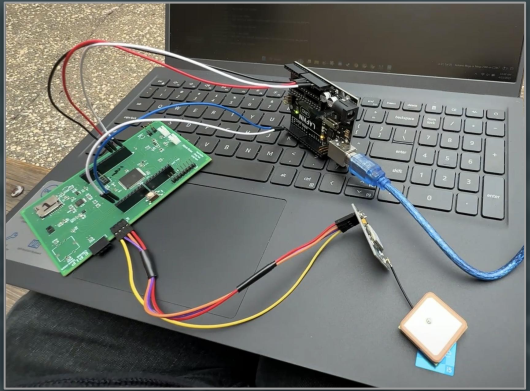


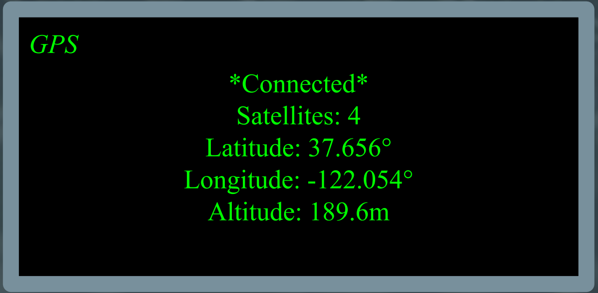
VDD RX1 TX1 GND

Dang T.

GPS: Milestones

Milestone 1: Works on PCB board
Milestone 2: Long initial cold start and limited satellite fixing
Milestone 3: Continuous uptime of GPS module makes for ideal efficient operation
Milestone 4: One limitation is the GPS antenna sensitivity; requires user to be outdoors or in open space to maintain reliable connection



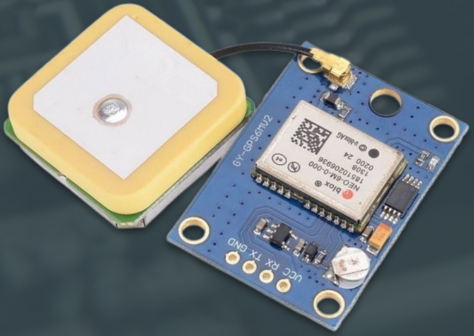


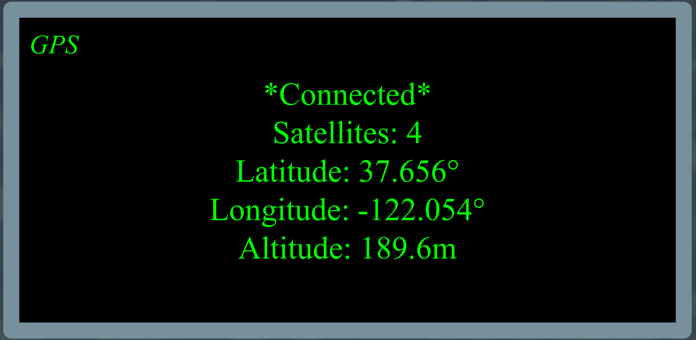
GPS Display Example

GPS: Summary

- NEO-6M module is key to providing accurate and reliable satellite-based location tracking;
- NEO-6M has its limitations, such as long initial start time & limited satellite acquisition;

- We use TinyGPSPlus library to properly integrate GPS functionality;
- Communication between NEO-6M and MCU is handled via UART;
- *Overall implementation aims to deliver a practical and dependable method for real-time location tracking*

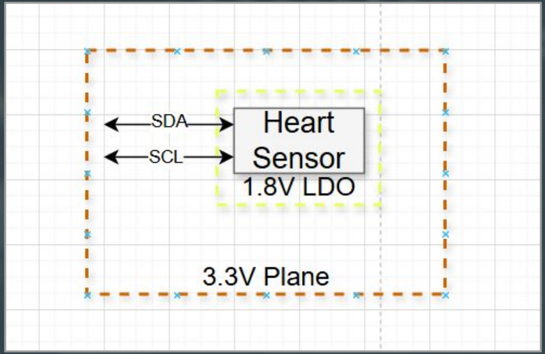




VI. Vitals

💓 Vitals: Overview

📝Requirements:	</> Software:	🔗Integration:	🚩 Reach Goals:
• 1.8V LDO powering sensor • 3.3V powering LEDs within module	• Arduino Software • SparkFun Particle Sensor Arduino Library	• Connection to MCU via I^2C	• Faster sensor readings • Cleaner sensor output



Noel A.

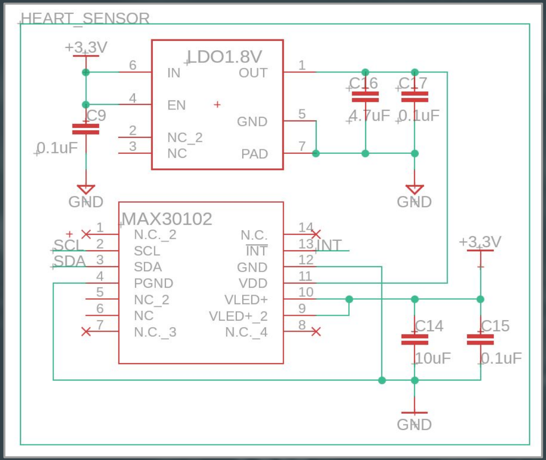
💓 Vitals: Schematic & Layout

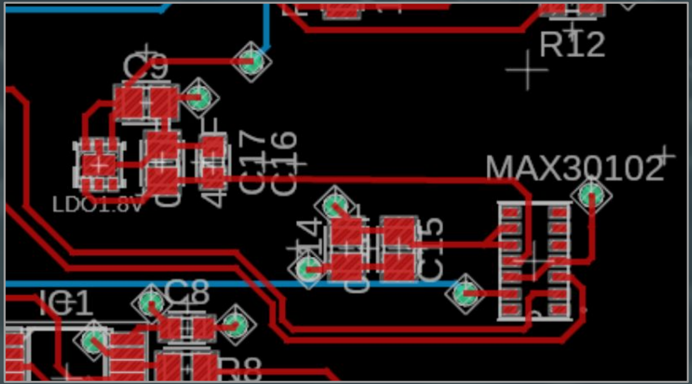
Schematic:

- Pin 2 & 3 Connects to MCU via I²C
- Pin 9 & 10 3.3V domain
- Pin 11 1.8V LDO
- Pin 12 GND

Layout:

- Center-Right of PCB
- Decoupling Capacitors on LDO and VLED





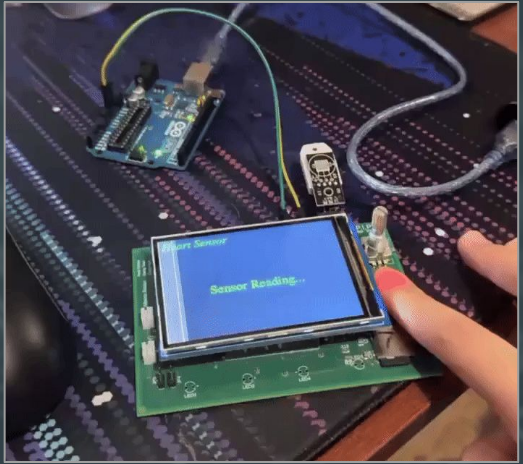
Noel A.

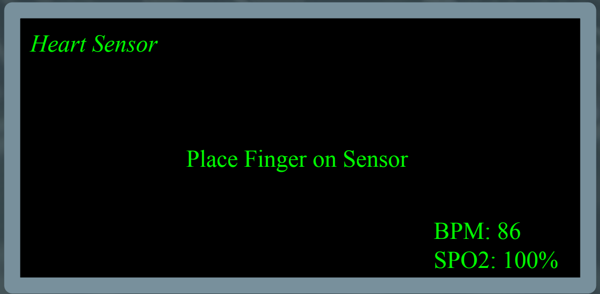
💓 Vitals: Milestones

Milestone 1: Verified all power domains (3.3V & 1.8V) and working LED control through I2C

Milestone 2: Sensor readings achieved allowing for heart rate and blood oxygen calculations

Milestone 3: BPM and SPO2 code implementation and fine tuning for better accuracy





Vitals Display Example

💾

VII. Storage

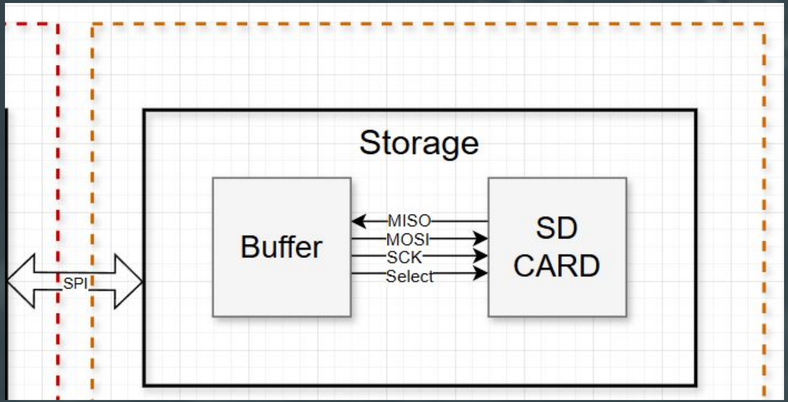
💾 Storage: Overview

📝Requirements:
- Log sensor alerts
- Timestamp entries
- Store user data

</> Software:
- Event-driven logging
- Filename versioning

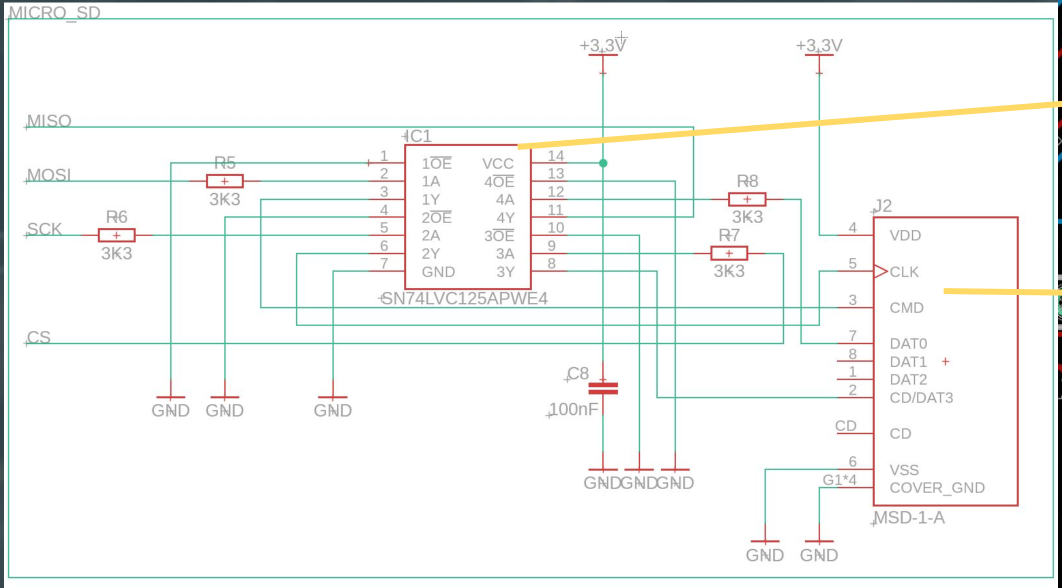
🔗Integration:
- Connect via SPI to MCU
- Used sensor flags for events
- Sync with GPS/Uptime

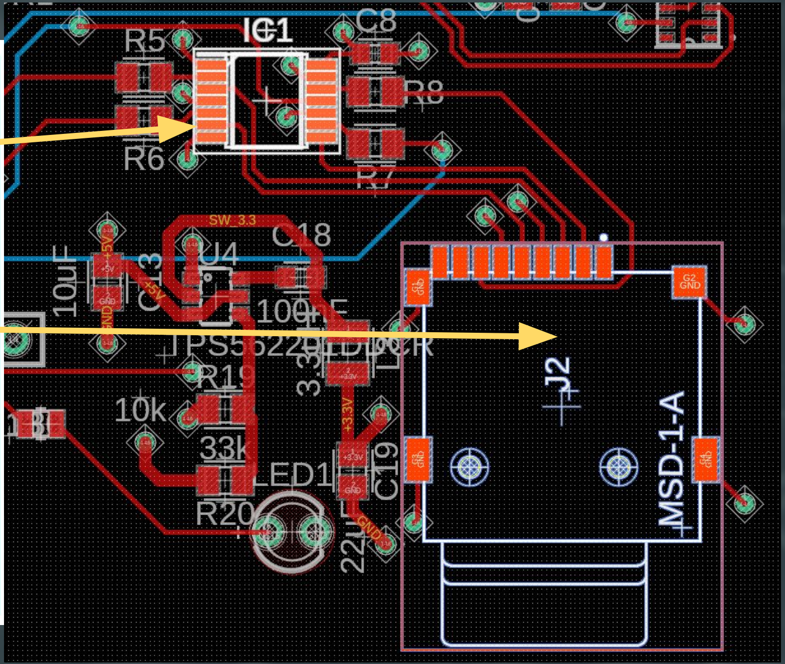
🚩 Reach Goals:
- On-screen log viewer
- Program to format Storage and add user file



Alberto R.

💾 Storage: Schematic & Layout

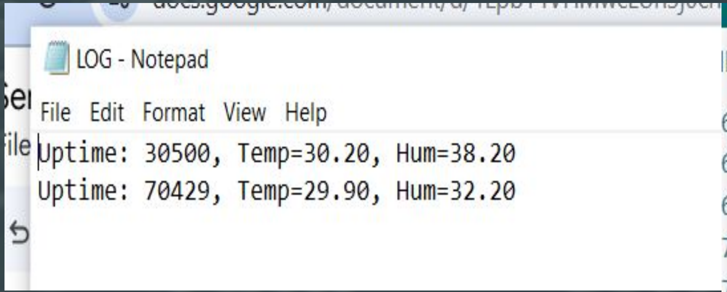


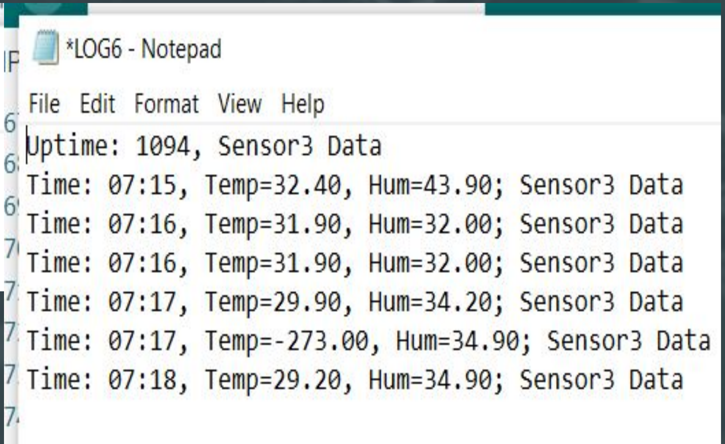


Alberto R.

💾 Storage: Milestones

Milestone 1: Power verification & card detection	
Milestone 2: Logging functionality and session numbering	



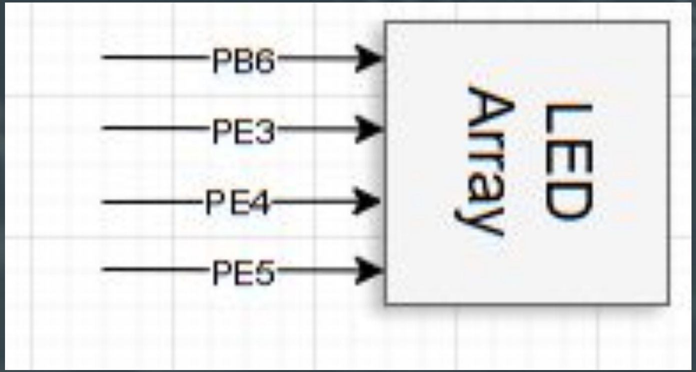


⌨️🚨

VIII.
Rotary Encoder & LED Array

🚨 LED Array: Overview

📝Requirements:	</> Software:	🔗Integration:	🚩 Reach Goals:
• Light up • Read stored data • Detect flag	• Check store data is flagged • Light up to indicate if over limit	• Soldered on Red LED to GPIO lines	• Integrate light fade to replicate blinking



Cody H.

⌨️ Rotary Encoder: Overview

📝Requirements:

- 5V power domain
- Switch multiple menus on display
- User input

</> Software:

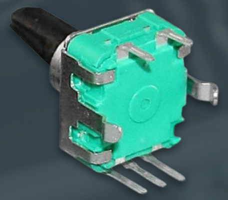
- State machine
- Debounce delay
- Idle state + Rotation modes

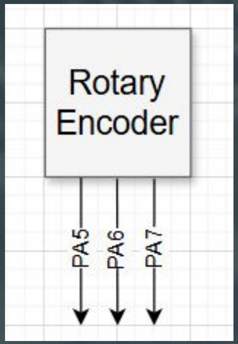
🔗Integration:

- MCU pins
- Pin 27: Button press
- Pin 28: -Rotation
- Pin 29: +Rotation

🚩 Reach Goals:

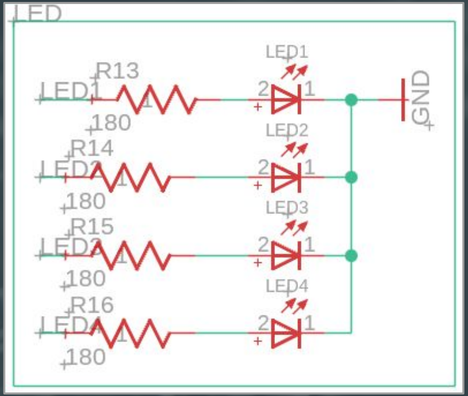
- Less false triggers

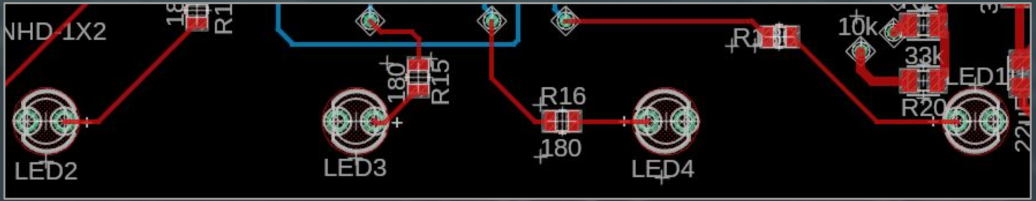


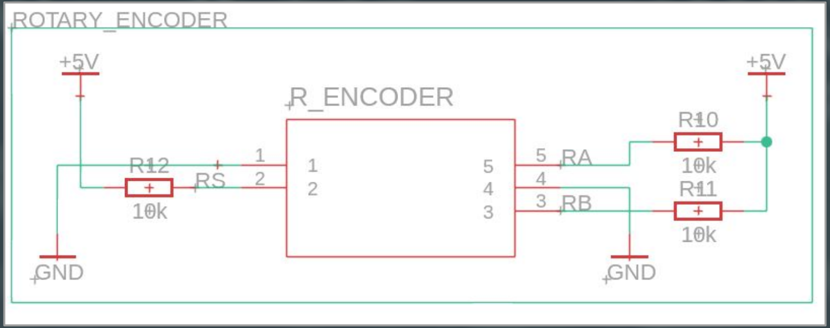


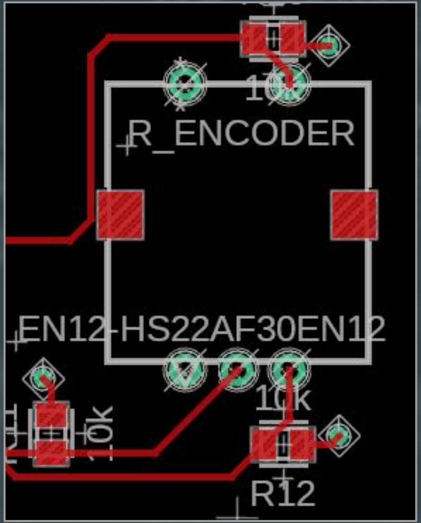
Cody H.

🚨 LED Array & ⌨️ Rotary Encoder: Schematic & Layout









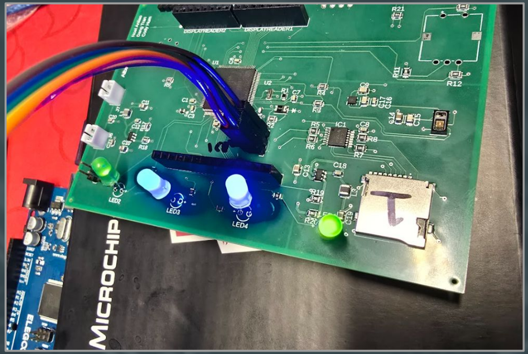
Cody H.

🚨 LED Array & ⌨️ Rotary Encoder: Milestones

Milestone 1: Power verification correct for both Rotary Encoder & LED Array, however input from RE was incorrect (CLK & GND).

- Corrected in software via GPIO mapping

Milestone 2: Mechanical bounce/Signal stability refined for Rotary Encoder. LED Array implementation for respective sensor display menus



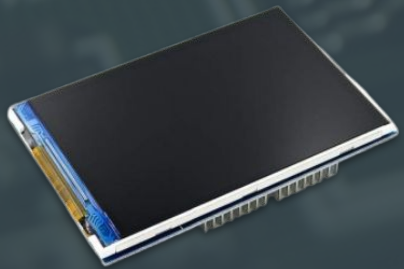

```
void input() {
  int currentStateCLK = digitalRead(ENCODER_CLK);
  // Check if the rotary encoder has turned
  if (currentStateCLK != lastStateCLK && currentStateCLK == HIGH && counter!=-1) {
    counter = (digitalRead(ENCODER_DT) != currentStateCLK) ? counter+1 : counter-1;

    // Loop counter from 0 to 3
    if (counter > 3) counter = 0;
    if (counter < 0) counter = 3;
  }
  // Check if the button is pressed
  if (digitalRead(ENCODER_SW) == LOW) {
    counter = (counter == -1) ? counter = 0 : counter = -1;
    delay(200);  // Simple debounce
  }
  lastStateCLK = currentStateCLK;
}
```

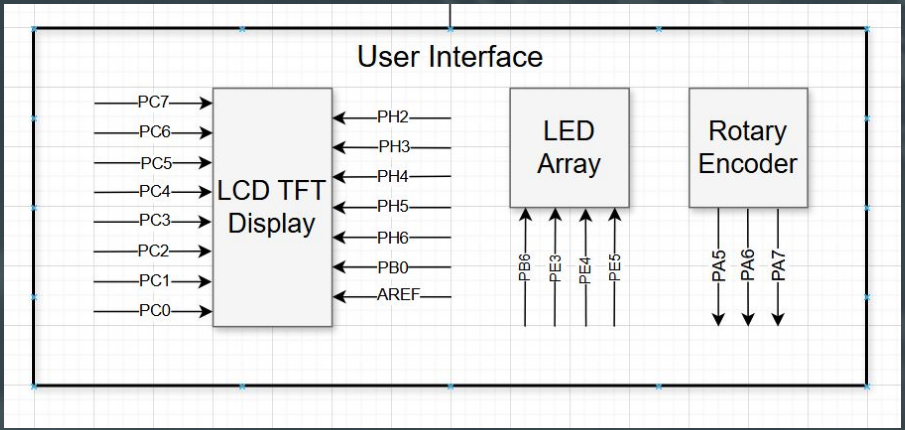
Cody H.

IX. Display

🖥️ Display: Overview

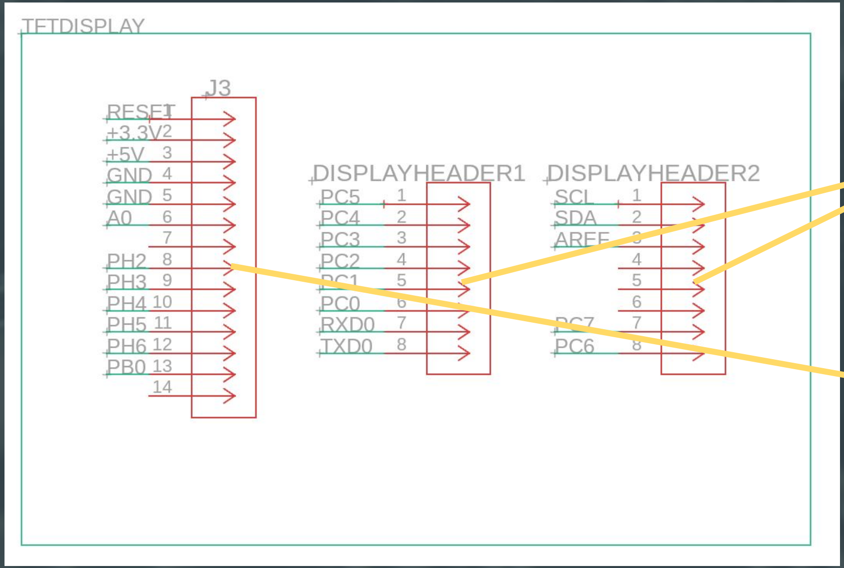
📝Requirements:	</> Software:	🔗Integration:	🚩 Reach Goals:
• Communication via GPIO headers on PCB • 5V power	• Adafruit_GFX library • MCUFRIEND_kbv library	• GPIO pin headers connected to the MCU (Parallel Interface)	• Faster GUI navigation • Fancier/cleaner GUI

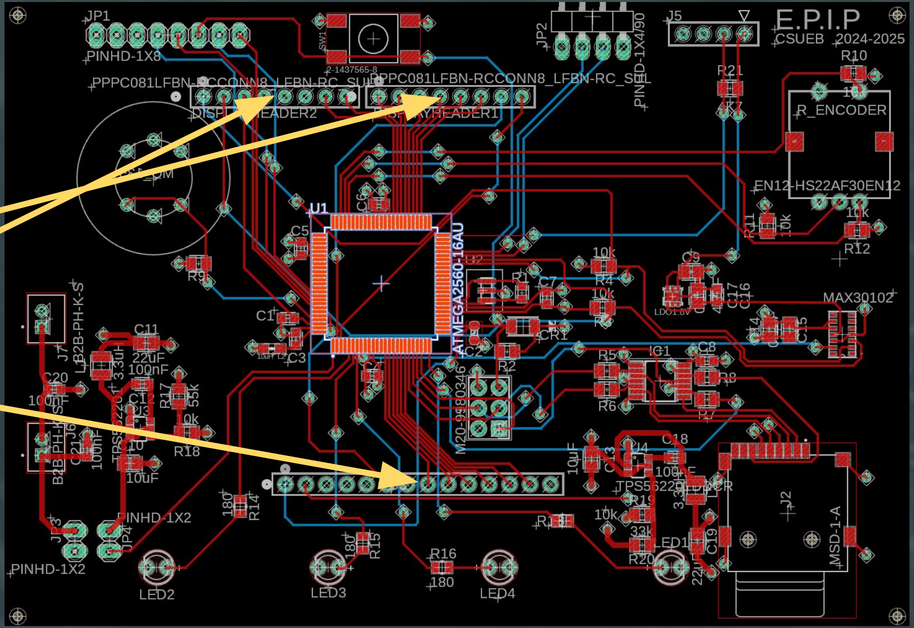




Noel A.

🖥️ Display: Schematic & Layout

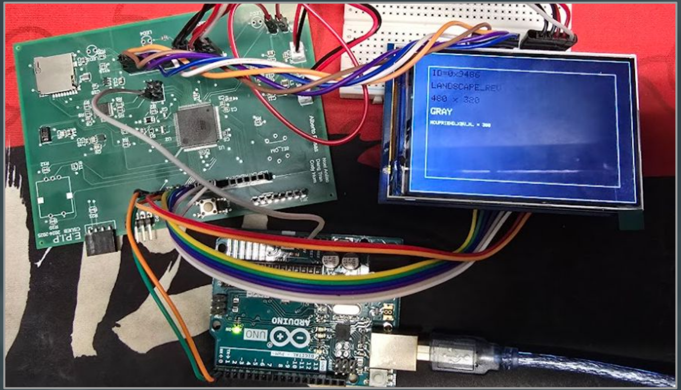


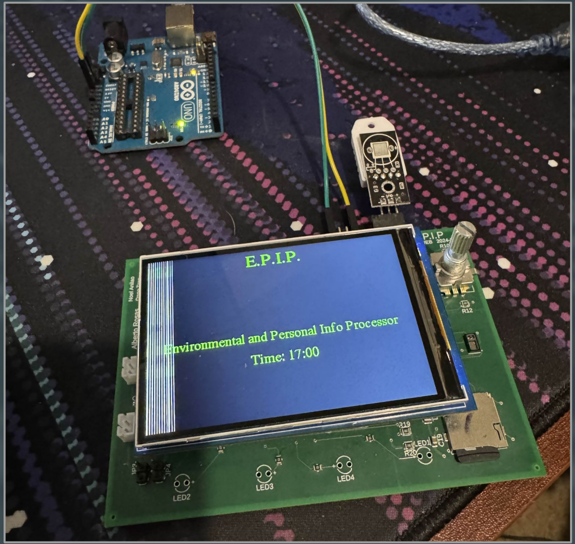


Noel A.

🖥️ Display: Milestones

Milestone 1: Verified Power domains, and correct display output via parallel pins
Milestone 2: Sensor integration and respective display menu outputs with data
Milestone 3: GUI refinement





Noel A.

X. Firmware

Firmware: Overview

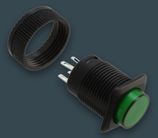
Startup: Initializes all sensors, LCD, and user interface, power-on sequence with green LED confirmation.
Sensor Management: Executes read functions for temperature/humidity, air quality, and GPS sensor every 3 sec in a non blocking loop. Vitals sensor is only read when user prompts it.
User Interface: Rotary encoder for screen navigation. Update screen by redrawing or updating data. LED alerts are triggered by relevant sensor flags.
Data Logging: Logs environmental and vital data to Micro SD card.

⚡ XI. Power Supply

⚡ Power Supply: Overview

📝Requirements:

- x2 3.7V lithium-ion batteries
- x2 buck-switching regulators
- Switch pushbutton ↓

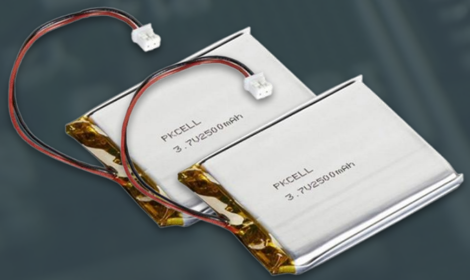


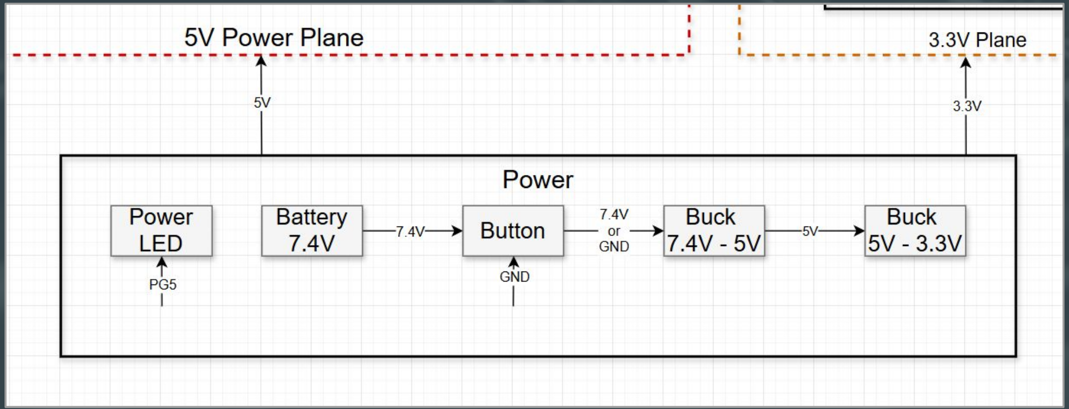
🔗Integration:

- Portable power supply
- Regulated via buck-switching regulators
- 5V and 3.3V domains

🚩 Reach Goals:

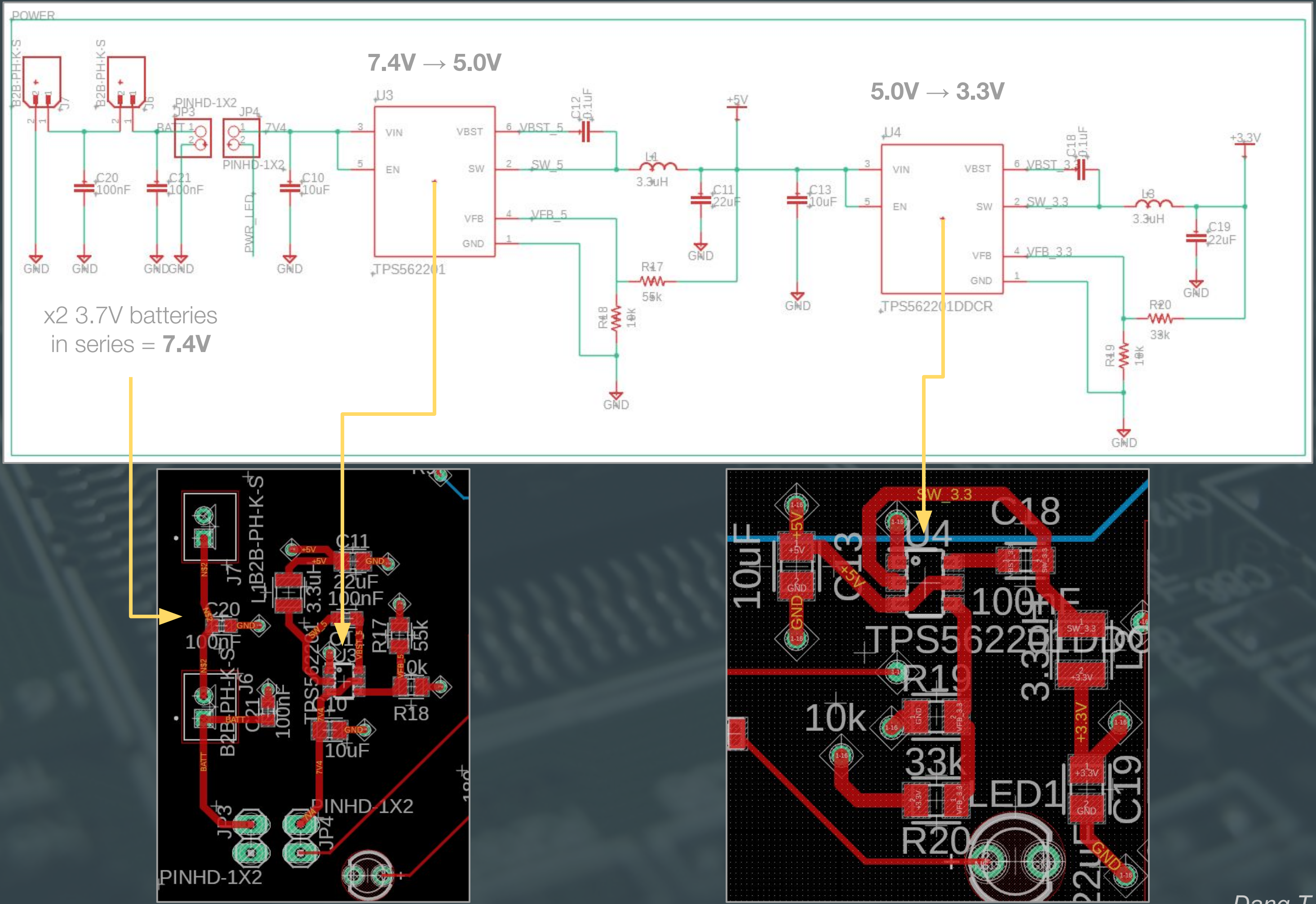
- Eliminate gradual voltage dissipation





Dang T.

Power Supply: Schematic & Layout



Dang T.

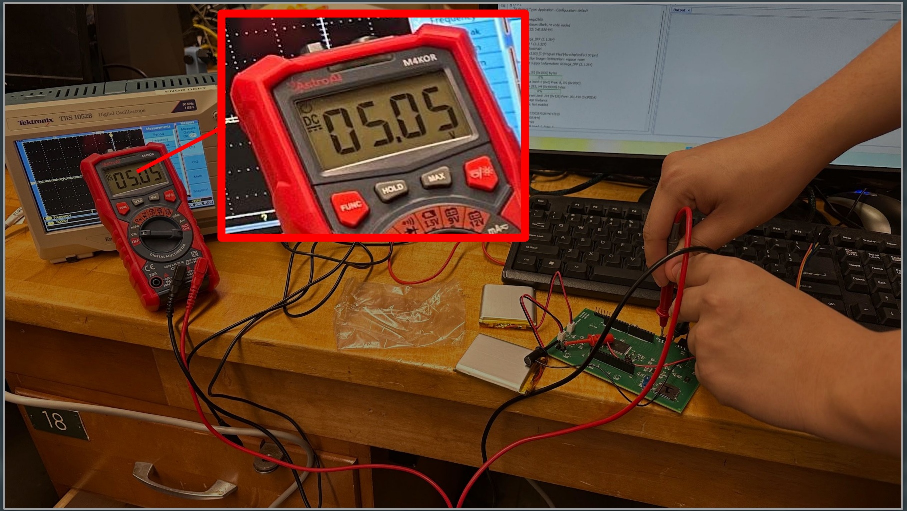
⚡ Power Supply: Milestones + Summary

🏔️ Milestones:

1. Successful confirmation on a working power supply; buck regulators step down to 5V and 3.3V
2. Minor residual voltage after powering down; gradually dissipation to 0.001V

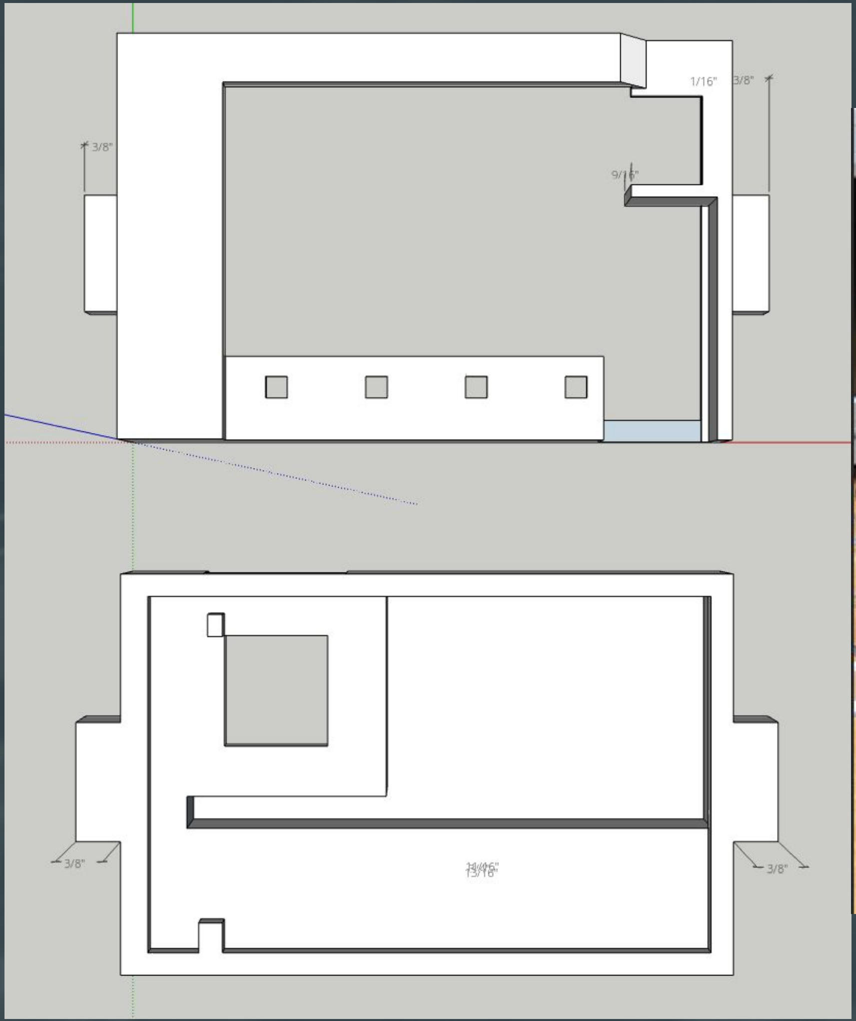
📝 Summary:

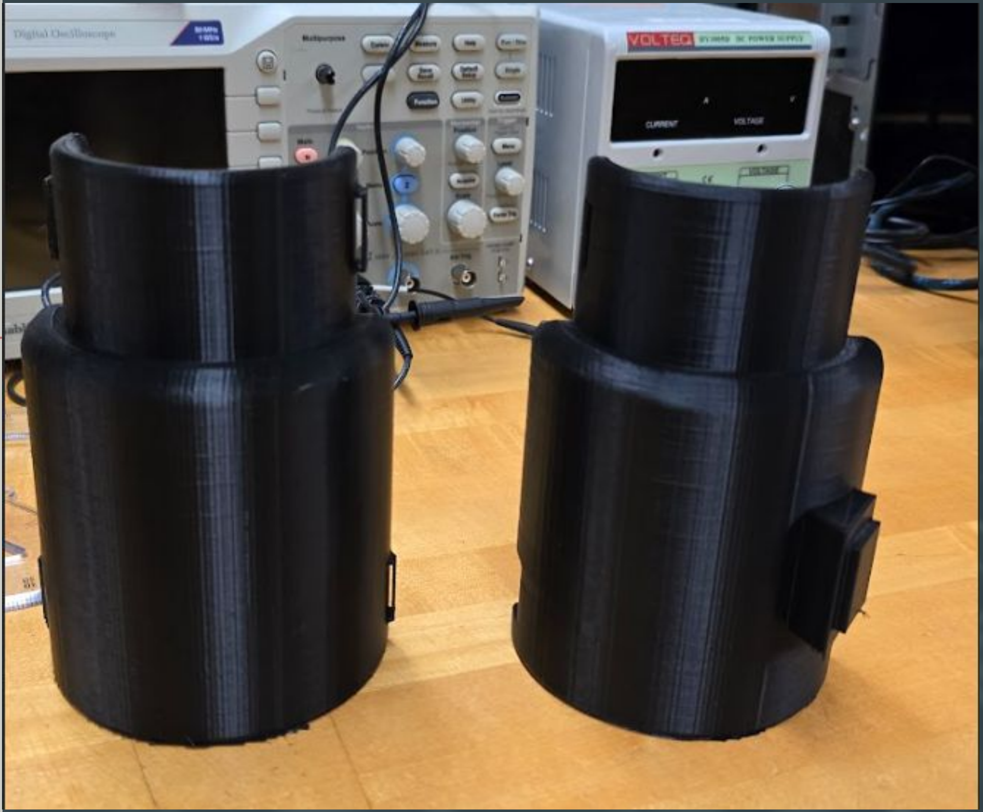
- Operates at 7.4V (x2 3.7V lithium-ion batteries in series)
- Two voltage domains, both regulated using buck-switching regulators
- Tiny issue in our layout design; not as efficient, but it works



Dang T.

XII.
CASING

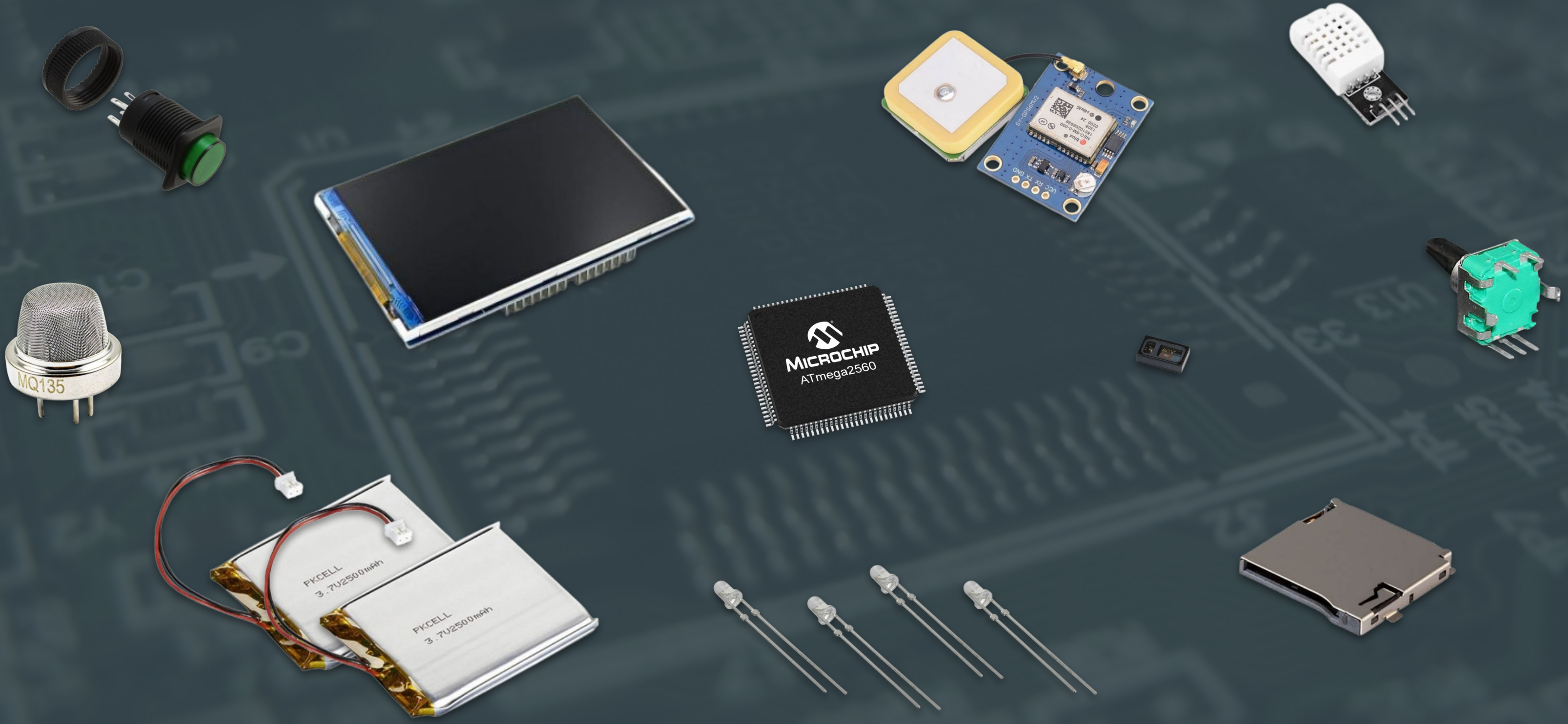




Alberto R.

XIII. CONCLUSION

Conclusion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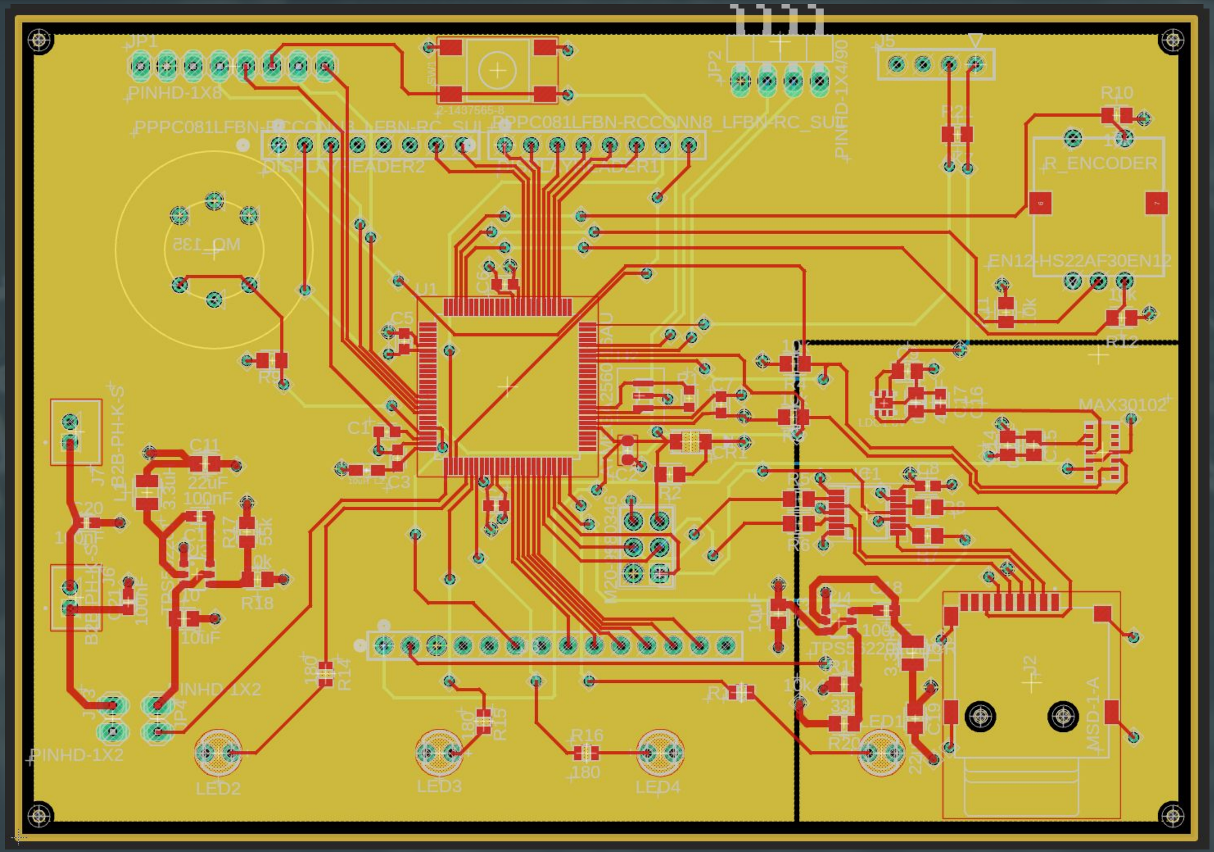
MICROCHIP
ATmega2560
PKCELL
3.7V2500mAh
PKCELL
3.7V2500mAh

Conclusion

Dang T.

Q&A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R12
U1
MAX30102
C11
22uF
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R18
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LED2
LED3
LED4
R16
180
J2
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